

SCN-Attention U-Net — Architecture Specification

PAUT Explainable-AI Digital Twin — weld defect inspection from Total Focusing Method (TFM) images.

Auto-generated by `scripts/model\_spec.py` from the production `configs/model.yaml`. Every number below is read from the instantiated model (forward-pass shapes + parameter counts), not hand-derived.

Input: $1 \times 1 \times 256 \times 256$ (single-channel amplitude patch)

Encoder (contracting path)

- **Depth:** 4 encoder stages + 1 bottleneck.
- **Widths (channels):** $32 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512$ (base = 32; each stage doubles).
- **Each stage:** $2 \times (\text{Conv}3 \times 3 \rightarrow \text{GroupNorm} \rightarrow \text{ReLU})$ (2 conv layers per stage $\rightarrow$ 8 conv layers in the encoder + 2 in the bottleneck).
- **Downsampling:** MaxPool2d(2) between stages ($256 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 16$ spatial).

Scattering branch (fixed, not trained)

- **Params:** $J=2, L=8, \text{order}=2$.
- **Coefficients:** 81 channels = 1 (order-0 low-pass) + 16 (order-1) + 64 (order-2).
- **Tensor:** $81 \times 64 \times 64$ (spatial = $\text{input} / 2^J = 256/4 = 64$).

Latent space (bottleneck)

- **Bottleneck tensor:** $512 \times 16 \times 16 = 131,072$ activations.
- **Pooled latent vector:** 512 (global-average-pool of the bottleneck — feeds the classifier).
- **Classifier input:** 512 (pooled latent) + 81 (global scattering vector) = 593 $\rightarrow$ Linear(128) $\rightarrow$ 2 classes.

Decoder (expanding path)

- $4 \times (\text{ConvTranspose}2d \times 2 \text{ up} \rightarrow \text{AttentionGate}(\text{skip}) \rightarrow \text{ConvBlock})$.
- **Skip connections** are filtered by additive Attention Gates (Oktay 2018) before concatenation.

Layer-by-layer table

Stage	Output (C×H×W)	Trainable params	Frozen params
scat	$81 \times 64 \times 64$	0	0
enc1	$32 \times 256 \times 256$	9,632	0
fuse1	$32 \times 256 \times 256$	5,126	0
enc2	$64 \times 128 \times 128$	55,552	0
fuse2	$64 \times 128 \times 128$	14,762	0
enc3	$128 \times 64 \times 64$	221,696	0
fuse3	$128 \times 64 \times 64$	47,858	0
enc4	$256 \times 32 \times 32$	885,760	0
fuse4	$256 \times 32 \times 32$	169,346	0

Stage	Output (C×H×W)	Trainable params	Frozen params
bottleneck	512×16×16	3,540,992	0
fuseb	512×16×16	633,506	0
up4	256×32×32	2,361,475	0
up3	128×64×64	590,915	0
up2	64×128×128	148,003	0
up1	32×256×256	37,139	0
seg_head	3×256×256	99	0
cls_head	2	76,290	0
TOTAL	—	8,798,151	0

Total parameters: 8,798,151 trainable. The Wavelet-Scattering filters are **fixed buffers, not learnable parameters** (Kymatio stores them as non-trainable buffers), so they contribute **0** to the parameter count — exactly the point of the SCN prior. Only the small 1×1 fuse* projection convs that adapt the scattering tensor to each U-Net scale are learned.

Hand-verification (math == code, sir's point 9)

A few stages checked by hand against the table above:

- **Scattering coeffs:** $1 + J \cdot L + L^2 \cdot J(J-1)/2 = 1 + 2 \cdot 8 + 8^2 \cdot 2 \cdot 1/2 = 81$. ✓
- **enc1 ConvBlock(1→32):** conv1 (1·32·3·3=288) + GN(2·32=64) + conv2 (32·32·3·3=9216) + GN(64) = **9,632**. ✓
- **seg_head Conv2d(32→3, 1×1):** 32·3 + 3 = **99**. ✓
- **cls_head:** Linear(593→128)=75,904+128 + Linear(128→2)=256+2 = **76,290**. ✓